

Project Design Phase

Solution Architecture

Date	2 August 2026	Team ID	Not specified
Project Name	Digitize Metro Ticket Booking Using ServiceNow	Maximum Marks	4 Marks

The proposed architecture uses ServiceNow platform capabilities to digitize the metro ticket booking lifecycle. The architecture connects the passenger booking interface with Service Catalog, automated Flow Designer processes, fare calculation, QR-code ticket generation, notifications, and ticket status management.

Architecture and Data Flow

Passenger Selects source, destination, travel details, and ticket quantity.	Service Catalog Collects structured metro ticket booking information.	Flow Designer Validates booking data and automates the ticket workflow.
Fare Calculation Calculates the applicable fare using journey details and configured fare rules.	Ticket Generation Creates the digital ticket and generates a unique QR code.	Notifications Sends booking confirmation and digital ticket details to the passenger.
Metro Entry Passenger presents the QR code for ticket verification and access.	Ticket Record Stores booking details, journey information, fare, and ticket status.	Administration Supports monitoring, reporting, and management of ticket transactions.

Component Description

Component	Function
Passenger / User	Initiates the booking by providing source station, destination station, travel details, and required ticket quantity.
Service Catalog	Provides the digital request interface and captures standardized metro ticket booking information.
Flow Designer	Automates the booking process, validates inputs, coordinates fare calculation, and triggers ticket generation and notifications.
Fare Calculation Logic	Determines the ticket fare based on configured journey and fare rules.
Ticket Record	Stores the booking information, fare, QR-code reference, ticket status, and other relevant details.
QR Code Generation	Creates a digital QR code associated with the generated ticket for verification at the metro entry point.

Notifications	Sends confirmation and ticket information to the passenger after successful ticket generation and can provide relevant status updates.
Metro Verification	Allows the digital QR code to be presented or scanned for ticket validation, subject to the metro operator's access-control system.
Reporting and Administration	Provides visibility into bookings, ticket status, travel patterns, and operational information.

End-to-End Process Flow

1. Passenger opens the digital metro ticket booking service. 2. Passenger selects the source station, destination station, travel details, and ticket quantity. 3. Service Catalog captures and submits the booking request. 4. Flow Designer validates the request and initiates the automated workflow. 5. Fare calculation logic determines the applicable fare. 6. The system creates the ticket record and generates a unique QR code. 7. Notifications send the booking confirmation and digital ticket information to the passenger. 8. Passenger presents the QR code at the metro entry point for verification. 9. Ticket status and booking information remain available for operational monitoring and administration.

Technology Stack

Platform: ServiceNow **Booking Interface:** Service Catalog **Automation:** Flow Designer **Communication:** ServiceNow Notifications **Data:** ServiceNow tables and ticket records **Digital Ticket:** QR-code based ticket representation.

Benefits

- Modular and easy to maintain
- Secure and role-based access
- Automated ticket booking workflow
- Fast QR-code ticket generation
- Centralized data management
- Improved passenger experience
- Reduced operational workload

Conclusion

The proposed Solution Architecture for the **Digitize Metro Ticket Booking Using ServiceNow** project provides a structured and scalable design that integrates user interaction, validation, workflow automation, business logic, secure data storage, QR-code generation, notifications, and reporting into a single platform. By leveraging ServiceNow's built-in capabilities such as the Service Catalog, Flow Designer, Business Rules, and custom tables, the system simplifies metro ticket booking, enhances passenger convenience, reduces manual effort, and enables efficient administration. This architecture also provides a strong foundation for future enhancements, ensuring that the solution remains adaptable to evolving transportation and digital service requirements.